MIL-P-24293A(SH)
19 August 1987
SUPERSEDING
MIL-P-24293(SHIPS)
25 April 1967
(See 6.4)

MILITARY SPECIFICATION

PACKING, HYDRAULIC, LEATHER, CATTLEHIDE, MINERAL-TANNED, IMPREGNATED

This specification is approved for use within the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers packing composed of impregnated, mineral tanned leather.

1.2 Classification. The leather packing shall be furnished in the following types and weights, as specified (see 6.2):

Type I - Impregnated with wax.
Type II - Impregnated with liquid polymer of polysulfide rubber.
Medium.
Heavy.
Extra heavy.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 5523, Department of the Navy, Washington, DC 20362-5101 by using the self-addressed Standardization Document Improvement Prposal (DD Form 1426) appearing at the end of this document or by letter.

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SPECIFICATIONS

FEDERAL

V-T-276 - Thread, Cotton.
QQ-S-781 - Strapping, Steel, and Seals.
UU-P-268 - Paper, Kraft, Wrapping.
CCC-C-467 - Cloth, Burlap, Jute (Or Kenaf).
PPP-B-1055 - Barrier Material, Waterproofed, Flexible.

STANDARDS

FEDERAL

FED-STD-311 - Leather, Methods of Sampling and Testing.
FED-STD-751 - Stitches, Seams, and Stitchings.

MILITARY

MIL-STD-129 - Marking for Shipment and Storage.

(Copies of specifications and standards required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

UNIFORM CLASSIFICATION COMMITTEE AGENT

Uniform Freight Classification Ratings, Rules and Regulations

(Application for copies should be addressed to the Uniform Classification Committee Agent, Tariff Publication Officer, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 Material. The packing shall be made of cattlehide leather.

3.2 Tannage. The leather packing shall be tanned with chromium salts without the use of vegetable tanning materials.

3.3 Finish. The leather packing shall be full-grain with the flesh side smooth and free from loose flesh. It may be lightly split or skived to facilitate impregnation.

3.4 Trim. The leather packing shall be in the form of belting butt bends as specified in FED-STD-311.

3.5 Selection. When provided in the form of belting butt bends, the leather packing shall have no defects such as deep cuts, damaged grain, open scratches, open grub holes, brands, hip marks, or any other defects which effect a loss in cutting value greater than 6 percent of the total area. When purchased in the form of finished packing, the leather packing shall be free from defects.

3.6 Thickness for weights. The leather packing shall be furnished in the thickness specified for each weight shown in table I.

TABLE I. Thickness for each weight of leather packing.

Weight	Thickness	
	Minimum	Maximum
	64ths inch	64ths inch
Medium	8	10
Heavy	11	13
Extra heavy	14	17

3.7 Cracking. The leather packing shall not crack (see 4.5).

3.8 Tensile strength. The tensile strength shall be not less than 2,500 pounds per square inch (lb/in^2) for any one specimen (see 4.5).

3.9 Elongation. The elongation under a load of 2,500 lb/in^2 shall be not more than 30 percent when tested with the long direction parallel to the backbone. When tested in the belly-belly direction, the elongation shall not exceed the following limits (see 4.5):

Medium - 30 percent maximum.
Heavy - 35 percent maximum.
Extra heavy - 40 percent maximum.

3.10 Shrinkage. The reduction in area of the leather packing shall be not more than 5 percent (see 4.5.2).

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3.11 Porosity. The impregnated leather packing shall show no leakage of water (see 4.5.3).

3.12 Chemical composition. The leather packing shall conform to the chemical composition shown in table II (see 4.5, 4.5.1, and 4.5.5).

TABLE II. Chemical composition.

	Percent by weight			
	Type I		Type II	
	Min	Max	Min	Max
Chromic oxide 1/	4.5	-----	4.5	-----
Grease (petroleum ether extract) 2/	8	20	8	15
Impregnating wax (benzol extract) 2/	23	30	-----	-----
Chlorides as sodium chloride 1/	-----	0.30	-----	0.30
Sulfates as sulfuric acid 1/	-----	0.30	-----	0.30
Total polysulfide polymer (Thiokol) 3/	-----	-----	35	-----
Chloroform extract 3/	-----	-----	-----	15

1/ Moisture, wax, and grease-free basis.
2/ Moisture-free basis.
3/ Determined on finished leather packing. Ash and all remaining chemical tests shall be determined on a representative sample prior to application of polysulfide polymer.

3.12.1 Ash (moisture, wax, and grease-free basis). Ash shall be not more than 120 percent of the chromic oxide (see 4.5).

3.12.2 Impregnating wax. The leather packing shall be impregnated with a wax having a melting point within the range of 170 to 185 degrees Fahrenheit (°F).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

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4.2 Sampling for quality conformance inspection.

4.2.1 Inspection lot. All leather packing of the same type and weight presented at one time shall be considered a lot for purposes of inspection.

4.2.2 Sample for examination. A random sample of leather packing units shall be selected in accordance with table III from each inspection lot of material offered for examination of visual and dimensional characteristics. Acceptance quality level (AQL) equals 1.5 percent defective.

TABLE III. Sampling for visual and dimensional examination.

Number of units in inspection lot	Number of units in sample	Acceptance number (defectives)	Rejection number (defectives)
15 and under	7	0	1
16 to 40	10	0	1
41 to 110	15	0	1
111 to 300	25	1	2
301 to 500	35	1	2
501 to 800	50	2	3
801 to 1300	75	3	4
1301 and over	110	4	5

4.3 Sampling for tests. Samples for all physical and chemical tests of type I packing and for all physical tests and for the determination of the polysulfide rubber and chloroform extract in type II packing shall be taken from the impregnated packing. Samples for the chromic oxide, chlorides, sulfates, and ash chemical tests on type II packing shall be taken from the unimpregnated packing representative of the lot. Sampling shall be in accordance with the sample size of sampling and acceptance procedures of FED-STD-311.

4.3.1 Test specimens. Specimens shall be cut from the sample units selected as specified in 4.3 in accordance with figure 1. Physical test specimens shall be cut as shown on figure 2 from piece "A" shown on figure 1. Specimens for chemical analysis shall be composited from pieces "B", "C_1", "C_2", and from remainder of "A" in accordance with method 6002.1 of FED-STD-311. Samples for chemical tests on unimpregnated type II packing shall be composited from pieces "B", "C_1", and "C_2".

4.4 Visual and dimensional examination. Each of the sample units selected in accordance with table III shall be visually and dimensionally examined to verify conformance to this specification. Any unit in the sample containing one or more visual or dimensional defects shall be subject to rejection and if the number of defective units in any sample exceeds the acceptance number for that sample, this shall be cause for rejection of the lot represented by the sample.

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4.5 Tests. Tests shall be conducted in accordance with the following methods as specified in FED-STD-311 and as hereinafter specified:

Test	Method
Moisture	6221
Grease (petroleum ether boiling below 80 degrees Celsius (°C))	6311
Chromic oxide, perchloric acid	6515
Soluble chlorides (on grease and wax-free sample)	6351
Soluble sulfates (on grease and wax-free sample)	6361
Total ash	6421
Tensile strength	2021
Elongation (at a stress of 2,500 lb/in^2)	2021
Cracking, vise	4011
Thickness, unit	1011.1

4.5.1 Impregnating wax (benzol extract). After completing the petroleum ether extraction the same packing and extract with benzol shall be used, using a Soxhlet extractor and keeping the benzol extraction separate from the petroleum ether extraction. The benzol shall be evaporated off and dried to approximately constant weight. This extraction shall be used for the melting point of the impregnating material.

4.5.2 Shrinkage. The length and width of the specimen shall be measured to the nearest 1/50 inch along the center lines of the specimen. The specimen shall be immersed in water at a temperature of 20 to 30°C for 24 hours and then heated to boiling at the rate of 5 to 7°C per minute and boiled for 6 minutes. The specimen shall be removed and the length and width again measured immediately. The shrinkage shall be computed as percentage of the original area.

4.5.3 Porosity. A circular porosity specimen, 3 inches in diameter made from each sample marked "A" shall support a head of 1 foot of water for 30 hours without showing porosity. For type II packing, Stoddard solvent shall be used in place of water.

4.5.4 Chloroform extract. Approximately 10 grams of type II packing shall be extracted with chloroform using method 6311 of FED-STD-311.

4.5.5 Total polysulfide polymer. Total polysulfide polymer shall be determined as follows. Accurately weigh 0.25 to 0.30 gram of impregnated packing and place in a Parr Bomb cup. Add approximately 0.50 gram of benzoic acid, 0.2 gram of potassium perchlorate and enough sodium peroxide to make the cup two-thirds full. Intimately mix the contents and ignite the bomb. When cool, dissolve the bomb contents in distilled water, filter, neutralize to methyl orange with hydrochloric acid then add a few drops in excess, bring solution to a boil and add 10 mL of 10 percent barium chloride with constant stirring. Let the solution stand for at least 16 hours and then filter. Wash the precipitate free of chlorides, then transfer the filter paper and precipitate to a tared crucible. Ignite to constant weight. The weight of barium sulfate is converted to percent sulfur by multiplying by 0.1373, dividing by the same weight and multiplying by 100. Repeat the determination on a representative sample of unimpregnated packing. Compute the percent of total polysulfide

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polymer content by subtracting the sulfur content of the unimpregnated sample from that of the impregnated and multiplying the difference by 2.74. (This factor is obtained by dividing 1/0.365 where 36.5 percent is representative of the sulfur content of the polysulfide polymer.)

4.6 Inspection of packaging. Sample packages and packs, and the inspection of the preservation-packaging, packing and marking for shipment and storage shall be in accordance with the requirements of section 5 and the documents specified therein.

5. PACKAGING

(The packaging requirements specified herein apply only for direct Government acquisition.)

5.1 Domestic shipment and early equipment installation.

5.1.1 Basic equipment or item.

5.1.1.1 Preservation and packaging. Preservation and packaging shall afford adequate protection against corrosion, deterioration and physical damage during shipment from the supply source to the using activity and until early installation and may conform to the contractor's commercial practice when such meets these requirements.

5.1.1.2 Packing. Packing shall be accomplished in a manner which will insure acceptance by common carrier at the lowest rate and will afford protection against physical or mechanical damage during direct shipment from the supply source to the using activity for early installation. The shipping containers or method of packing shall conform to the Uniform Freight Classification Rules and Regulations or other carrier regulations as applicable to the mode of transportation and may conform to the contractor's commercial practice when such meets these requirements.

5.1.1.3 Marking. Shipment marking information shall be provided on interior packages and exterior shipping containers in accordance with the contractor's commercial practice. The information shall include nomenclature, National stock number or manufacturer's part number, contract or order number, contractor's name and destination.

5.2 Domestic shipment and storage or overseas shipment. The requirements and levels of packaging, packing, and marking for shipment shall be as specified (see 6.2).

5.2.1 The following provides various levels of protection during domestic shipment and storage or overseas shipment, which may be required when acquisition is made.

5.2.1.1 Packaging. Packaging shall be level A or C, as specified (see 6.2).

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5.2.1.1.1 Level A. Twenty-four or 36 leather packings, as specified (see 6.2), shall be packaged in rolls as follows:

5.2.1.1.1.1 Rolls. Each set of six leather packings shall be reversed so that the heads and butts, backbones and bellies are adjacent to the next set, and then neatly and tightly rolled together and wrapped in kraft wrapping paper of not less than 80 pounds basis weight (24 by 36-500) conforming to grade B, no. 2 of UU-P-268. The paper shall be folded to enclose each end of the roll completely and shall be secured with gummed kraft tape.

5.2.1.1.1.1.1 Overwrapping of rolls. In addition, each roll of leather packing shall be protected by overwrapping in one of the following materials:

(a) Single-faced corrugated fiberboard in which the corrugated paper is not less than 0.009-pound sulfate kraft and the linerboard is of not not less than 47 pounds bursting strength (Mullen or Cady).
(b) Solid fiberboard of not less than 0.030-point caliper. The wrap shall extend not less than 1 inch beyond the ends of the roll and shall completely encircle the roll, overlapping 4 inches. The wrap shall be secured in place by tying with cord twice around the body approximately 12 inches from each end, and once lengthwise, tying with nonslip knots.

5.2.1.1.2 Level C. Packaging shall afford adequate protection against deterioration and physical damage during shipment from the supply source to the first receiving activity for early use and may conform to the contractor's commercial practice.

5.2.1.1 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1.2.1 Level A.

5.2.1.2.1.1 Bales. Rolls of leather packing shall be packaged as specified in 5.2.1.1.1, and packed in bales as specified in 5.2.1.2.2.1, except that in addition each roll shall be completely encased in an overwrap of waterproofed kraft wrapping paper conforming to class B-2 of PPP-B-1055, and tightly folded to enclose each end, prior to inserting the roll into cloth-covering material.

5.2.1.2.1.1.1. Strapping of bales. Flat nailless, painted steel straps, 5/8 inch wide by 0.018 inch thick, conforming to finish A of QQ-S-781 shall be used for binding the bales. Straps shall be applied around the bale spaced not less than 10 inches from each end, avoiding knots of cord used for inner wrapping. Tension shall be applied to the flat-metal strapping to insure tight application.

5.2.1.2.2 Level B.

5.2.1.2.2.1 Bales. Rolls of leather packings, packaged as specified in 5.2.1.1.1, shall be inserted into cloth covering material which has been formed into bias-sewn tubing, and the tubing shall be pulled tight around all sides. Material used shall be jute burlap, 40-inch 10 ounces per linear yard, conforming

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to CCC-C-467. Bias-sewn tubing shall be made by placing the selvage edge of the cloth covering material to the opposite edge and stitching not less than 3/8 inch nor more than 3/4 inch from the edge; stitch type 401 and seam type SSa-1 or LSA-1, conforming to FED-STD-751, not less than three stitches per inch, not less than 12/4 thread in needle, and not less than 16/3 thread in looper shall be used. Proper widths of material shall be used so that the finished width of tubing shall fit the rolls of leather packings snugly. The degree of bias shall be not less than 40 nor more than 50, and the angle used within this range shall remain in a reasonable constant throughout the sewing operation. Thread shall conform to type 1A1 of V-T-276.

5.2.1.2.2.1.1. Sealing of bales. Sealing of the tubular bales shall be by one of the following methods:

(a) Bales may be sealed at each end with two double-loop, 6-inch, 16-gauge, galvanized wire ties. The first wire tie shall be applied as close to the bale as possible so there will be a minimum of slack in the tube, the cloth being gathered evenly to form an ear, and all edges of the cloth being 3 to 5 inches from the wire tie at the base of the ear. The second wire tie shall be applied at a distance approximately 1 inch from the first wire tie. The twisted ends of the wire ties shall alternate and face in opposite directions.
(b) Bales may be sealed with one end presewn flat, seam to be 1 inch from end of precut tube, the other end to be closed with two wire ties.
(c) Handsewn bales will be permitted, if specified by the Command or agency concerned.

5.2.1.2.3 Level C. Leather packing, packaged as specified in 5.2.1.1.1, shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. Containers shall be in accordance with the Uniform Freight Classification Rates, Rules and Regulations or other carrier regulations as applicable to the mode of transportation and may be the contractor's commercial practice when such meets the requirements of this level.

5.2.1.3 Marking.

5.2.1.3.1 Interior packages. Each interior package shall be durably and legibly marked with the following information in such a manner that the markings will not become damaged when the packages are opened:

Packing, Hydraulic, Leather, Cattlehide, Mineral-Tanned, Impregnated.
Specification MIL-P-24293.
Type
Selection
Weight (medium, heavy, or extra heavy)
Order No.
Quantity
Manufacturer's name or trade-mark
Supplier's name (if not same as manufacturer's)

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5.2.1.3.2 Packages and containers. In addition to any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with MIL-STD-129. The nomenclature shall be "Packing, Hydraulic, Leather, Cattlehide, Mineral-Tanned, Impregnated".

6. NOTES

6.1 Intended use. This material is intended primarily for hydraulic leather packing furnished in the form of belting butt bends, but may be ordered as finished packing. Type I is intended for use with hydraulic pistons operating at pressures up to 3,000 lb/in^2 at temperatures up to 155°F. Type II is intended for use with oil pressures up to 5,000 lb/in^2 in the temperature range from minus 65 to plus 200°F.

6.2 Ordering data. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type and weight required (see 1.2).
(c) Number of leather packings packaged in a roll (see 5.2.1.1.1).
(d) Packaging, packing or marking requirements (see 5.2.1.1 and 5.2.1.2).

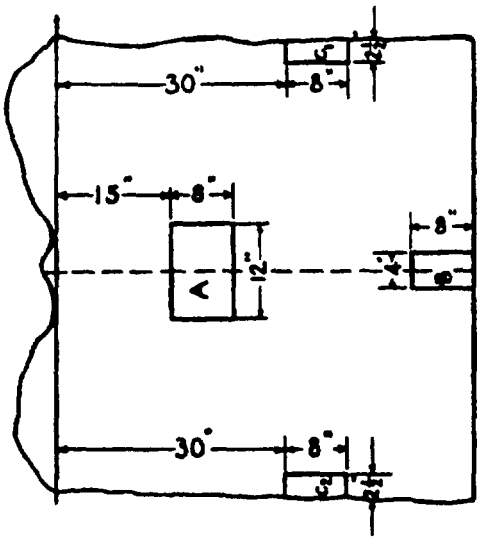
6.3 Subject term (key word) listing.

Belting butt bends
Chromium salts
Full-grain
Sulfate

6.4 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

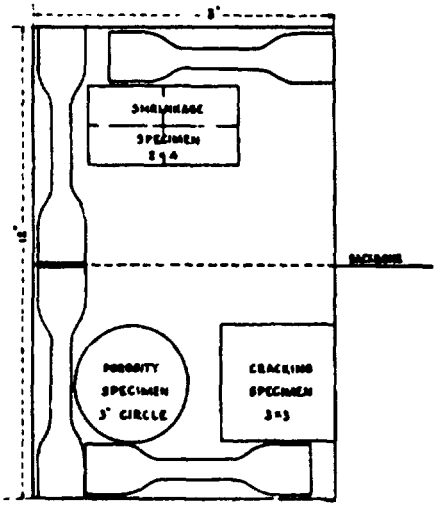
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SH 132317418

FIGURE 1. Location from which test specimens are to be taken.



SH 132317419

FIGURE 2. Specimen A - layout recommended.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1 DOCUMENT NUMBER	2 DOCUMENT TITLE
MIL-P-24293A(SH)	PACKING, HYDRAULIC, LEATHER, CATTLEHIDE, MINERAL TANNED, IMPREGNATED

3a. NAME OF SUBMITTING ORGANIZATION	4 TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)* _____
b ADDRESS *(Street, City, State, ZIP Code)*	

5 PROBLEM AREAS

a. Paragraph Number and Wording

b. Recommended Wording

c. Reason/Rationale for Recommendation

6 REMARKS

7a NAME OF SUBMITTER *(Last, First, MI)* – Optional	b WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8 DATE OF SUBMISSION *(YYMMDD)*

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NOTICE OF VALIDATION

MIL-P-24293A(SH)
NOTICE 1
27 March 1991

MILITARY SPECIFICATION

PACKING, HYDRAULIC, LEATHER, CATTLEHIDE, MINERAL-TANNED, IMPREGNATED

MIL-P-24293A(SH), dated 19 August 1987, has been reviewed and determined to be valid for use in acquisition.

Preparing activity:
Navy - SH

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